

Volatility in Exoplanet Habitability Classification Under Measurement Uncertainty

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Abstract

This study examines the stability of exoplanet habitability classifications under measurement uncertainty using data from the NASA Exoplanet Archive. While prior research has focused primarily on improving the accuracy of habitability prediction models, less attention has been given to the reliability of these predictions when input variables are subject to observational noise. To address this gap, this paper introduces a volatility-based framework that quantifies the sensitivity of classification outcomes to small perturbations in planetary and stellar features. A proxy habitability label is constructed using equilibrium temperature and planetary radius, and baseline classification models are trained to predict this label. Monte Carlo perturbations are then applied to evaluate the frequency of classification changes for each exoplanet. The resulting volatility score provides a measurement of prediction stability, highlighting planets whose classifications are highly sensitive to input uncertainty. The findings suggest that traditional performance metrics, such as accuracy, may fail to capture the reliability of model predictions in highly imbalanced datasets, as models with high overall accuracy may still perform poorly on rare habitable cases. In contrast, volatility analysis reveals that certain planets exhibit substantial classification instability despite stable baseline predictions, particularly those near model decision boundaries. These results demonstrate the importance of incorporating uncertainty into classification frameworks and suggest that prediction stability provides a valuable complement to standard evaluation metrics in exoplanet habitability analysis.

The discovery and classification of exoplanets has become a central focus of modern astrophysics, with particular attention given to identifying planets that may support life. Advances in observational technologies and large-scale astronomical databases have enabled researchers to analyze thousands of confirmed exoplanets, often using machine learning techniques to classify planets based on their potential habitability. These approaches typically focus on predictive accuracy, seeking to determine whether a given exoplanet meets certain criteria associated with habitable conditions. However, habitability classification is inherently uncertain. Many of the features used in these models, such as equilibrium temperature and planetary radius, are derived from indirect observations and are subject to measurement error. Despite this, most classification frameworks treat input values as fixed and deterministic, potentially overlooking the impact of uncertainty on prediction outcomes.

This study shifts the focus from prediction accuracy to prediction stability. Specifically, it investigates how sensitive habitability classifications are to small perturbations in input features. Rather than asking whether a planet is classified as habitable, this paper examines how consistently that classification is maintained when realistic variations in measured data are introduced. To address this question, this study introduces a volatility-based framework for evaluating classification robustness. A volatility score is defined as the proportion of classification changes observed under repeated perturbations of input variables. This approach provides a simple and interpretable measure of prediction stability, allowing for the identification of exoplanets whose classifications are highly sensitive to measurement uncertainty.

By focusing on stability rather than solely accuracy, this work contributes a complementary perspective to existing exoplanet classification research. More broadly, it

highlights the importance of incorporating uncertainty into data-driven decision-making processes, particularly in domains where measurements are inherently noisy.

Literature Review

Machine learning has been widely applied to the classification of exoplanets and the identification of potentially habitable candidates, with studies employing techniques such as logistic regression, decision trees, and neural networks (Shallue & Vanderburg, 2018; Armstrong et al., 2017). These approaches typically rely on datasets such as those provided by the NASA Exoplanet Archive, which constrain planetary and stellar characteristics derived from observational data.

Prior research has primarily focused on improving classification performance, often through the use of more complex models or feature engineering techniques. While these methods have demonstrated success in identifying patterns within exoplanet data, they generally assume that input features are measured without error. In practice, however, astronomical measurements are subject to uncertainty due to limitations in observational instruments and indirect estimation methods (Kopparapu et al., 2013).

Some studies have incorporated uncertainty through probabilistic or Bayesian modeling approaches, particularly in astrophysical simulations and atmospheric modeling (Gregory, 2005; Lored, 2012). In machine learning contexts, uncertainty has also been explored through probabilistic frameworks such as Bayesian neural networks (Kendall & Gal, 2017). However, these methods are often highly technical and computationally intensive, limiting their accessibility and interpretability in broader data science contexts.

In contrast, there is relatively limited work examining the stability of classification outcomes under small perturbations in input data, particularly using simple and interpretable

models. This study addresses this gap by introducing a volatility-based framework that quantifies classification sensitivity in a straightforward and computationally efficient manner. By combining standard classification techniques with Monte Carlo-style perturbations, this approach provides a practical method for assessing the robustness of habitability predictions without requiring complex probabilistic modeling.

Method

Data Source

The dataset used in this study was obtained from the NASA Exoplanet Archive, which provides a comprehensive collection of confirmed exoplanet observations. Specifically, the Planetary Systems (PS) table was used, containing planetary and stellar attributes relevant to habitability analysis. The dataset includes variables such as planetary radius, orbital period, equilibrium temperature, and host star characteristics derived from observational measurements.

Data Preprocessing

The raw dataset required preprocessing due to formatting inconsistencies and missing values. Column names were assigned, and only variables relevant to habitability classification were retained. Observations with missing values in key features were removed to ensure consistency across modeling procedures. While uncertainty bounds were available for many variables, observations with missing uncertainty values were retained. In such cases, uncertainty was treated as zero, allowing these observations to be included in both deterministic and uncertainty-aware analyses.

Additionally, the dataset included multiple parameter sets per planet derived from different observational studies. To ensure consistency and avoid redundant representations, only the default parameter set provided by the archive was retained for each planet. These default

values represent the most reliable and recommended measurements based on current scientific consensus. After filtering, duplicate planet entries were removed, resulting in a single observation per planet.

In addition to central parameter estimates, the dataset includes asymmetric uncertainty bounds for many variables. These were retained and used to derive uncertainty-aware features, enabling downstream modeling procedures to incorporate measurement variability directly rather than treating observations as fixed values.

The final dataset included the following variables: planetary radius, orbital period, equilibrium temperature, stellar temperature, stellar radius, and stellar mass. These features were selected based on their relevance to planetary conditions and their availability across observations.

After preprocessing, the final dataset consisted of 1,483 observations, of which 9 were classified as proxy-habitable. This distribution reflects the substantial class imbalance present in the dataset and aligns with the rarity of habitable planet candidates in observational data. This imbalance was intentionally preserved to maintain consistency with the real-world distribution of observed exoplanets.

The code used for data preprocessing, modeling, and volatility analysis is available at: <https://github.com/carlycarroll25/Volatility-in-Exoplanet-Habitability-Classification-Under-Measurement-Uncertainty>.

Proxy Habitability Definition

Because true habitability cannot be directly observed from the available dataset, a proxy habitability label was constructed using simplified astrophysical criteria. A planet was classified

as proxy-habitable if it satisfied both of the following conditions: planetary radius less than or equal to 1.6 Earth radii and equilibrium temperature between 180 K and 310 K.

This definition provides a consistent and reproducible classification target while acknowledging that habitability is a complex and multifactorial concept not fully captured by these variables.

Modeling Approach

Two classification models were used to predict proxy habitability: logistic regression and random forest. Logistic regression was selected as a baseline model due to its interpretability and ability to provide insight into the relationship between input features and classification outcomes. Random forest was included as a nonlinear model capable of capturing more complex interactions among features.

Using both models allows for comparison between a simple, interpretable approach and a more flexible, ensemble-based method. This comparison also supports evaluation of whether classification stability varies across model types.

Model performance was evaluated using standard classification metrics, including accuracy, precision, recall, and F1 score. A 70/30 train-test split was used, resulting in 1,038 observations in the training set and 445 observations in the test set.

Uncertainty and Volatility Analysis

Measurement uncertainty was explicitly incorporated using the asymmetric error bounds provided in the dataset. For each numeric feature, the reported upper and lower uncertainties were used to approximate a characteristic uncertainty scale, enabling the construction of feature-specific normal distributions centered at the observed values with variance determined by the derived uncertainty scale. Rather than applying arbitrary noise, perturbations were generated

by sampling from these distributions, ensuring that simulated variability reflected realistic observational uncertainty.

While uncertainty bounds were available for many variables, observations with missing uncertainty values were retained. For these observations, no additional variability was introduced during perturbation, allowing them to be included in both deterministic and uncertainty-aware analyses.

For each planetary observation, Monte Carlo sampling was performed by repeatedly drawing perturbed feature values from these uncertainty-informed distributions. These samples were then passed through the trained classification model to evaluate the stability of predicted habitability under realistic measurement noise.

This approach leverages the inherent uncertainty structure of the NASA Exoplanet Archive, allowing for a more faithful representation of astrophysical measurement variability compared to traditional noise injection methods.

A volatility score was defined as the proportion of perturbed samples for which the predicted classification differed from the original classification. Formally, for a given observation i , the volatility score is defined as:

$$V_i = \frac{1}{M} \sum_{m=1}^M \mathbf{1} \left(\hat{y}_i^{(m)} \neq \hat{y}_i^{(0)} \right)$$

where M is the number of perturbations, $\hat{y}_i^{(0)}$ is the original predicted label, and $\hat{y}_i^{(m)}$ is the predicted label under the m -th perturbation.

This measure captures the sensitivity of classification outcomes to small changes in input data, providing an interpretable metric for assessing prediction stability.

Results

Two classification models, logistic regression and random forest, were trained to predict the proxy habitability label using planetary and stellar features. Model performance was evaluated on a held-out test set using accuracy, precision, recall, and F1 score.

Model Performance

The performance of logistic regression and random forest models are summarized in Table 1. Logistic regression achieved perfect recall for the proxy-habitable class but exhibited low precision, indicating a tendency to overpredict positive cases. In contrast, the random forest model achieved high overall accuracy and strong precision, along with moderate recall, indicating that the model was able to correctly identify positive cases while maintaining a low false positive rate.

These differences in model behavior reflect a tradeoff between sensitivity and precision. Logistic regression favors detection of rare positive cases at the cost of increased false positives, whereas the random forest model produces more conservative predictions with higher precision but lower recall. These results highlight the importance of evaluating multiple performance metrics when working with highly imbalanced datasets, as accuracy alone may not fully capture model effectiveness. The performance metrics for both logistic regression and random forest models are presented in Table 1.

Table 1

Model Performance Metrics

Model	Accuracy	Precision	Recall	F1 Score
Logistic Regression	0.935	0.094	1.000	0.171
Random Forest	0.998	1.000	0.667	0.800

Volatility Analysis

To assess prediction stability, Monte Carlo perturbations were applied to each observation in the test set using uncertainty-informed sampling. For each planet, a volatility score was computed as the proportion of perturbed predictions differing from the original predicted label.

As shown in Figure 1 and Figure 2, most observations exhibited low volatility, indicating stable predictions under small perturbations. However, a subset of planets showed significantly higher volatility, suggesting that their classifications are sensitive to measurement uncertainty.

Notably, certain planets displayed substantial instability despite having consistent baseline predictions. For example, some observations exhibited volatility scores above 0.4 under the random forest model, indicating nearly half of the perturbed samples changed classification.

Figure 1

Distribution of Volatility Scores for Logistic Regression

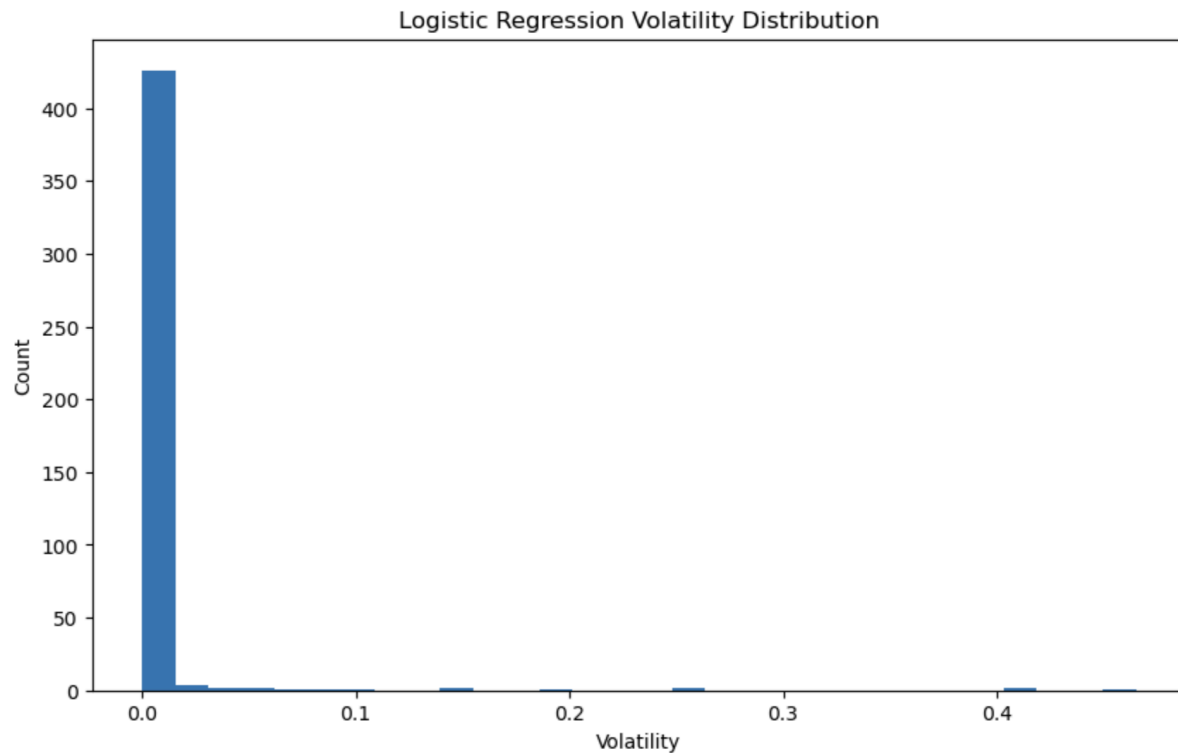
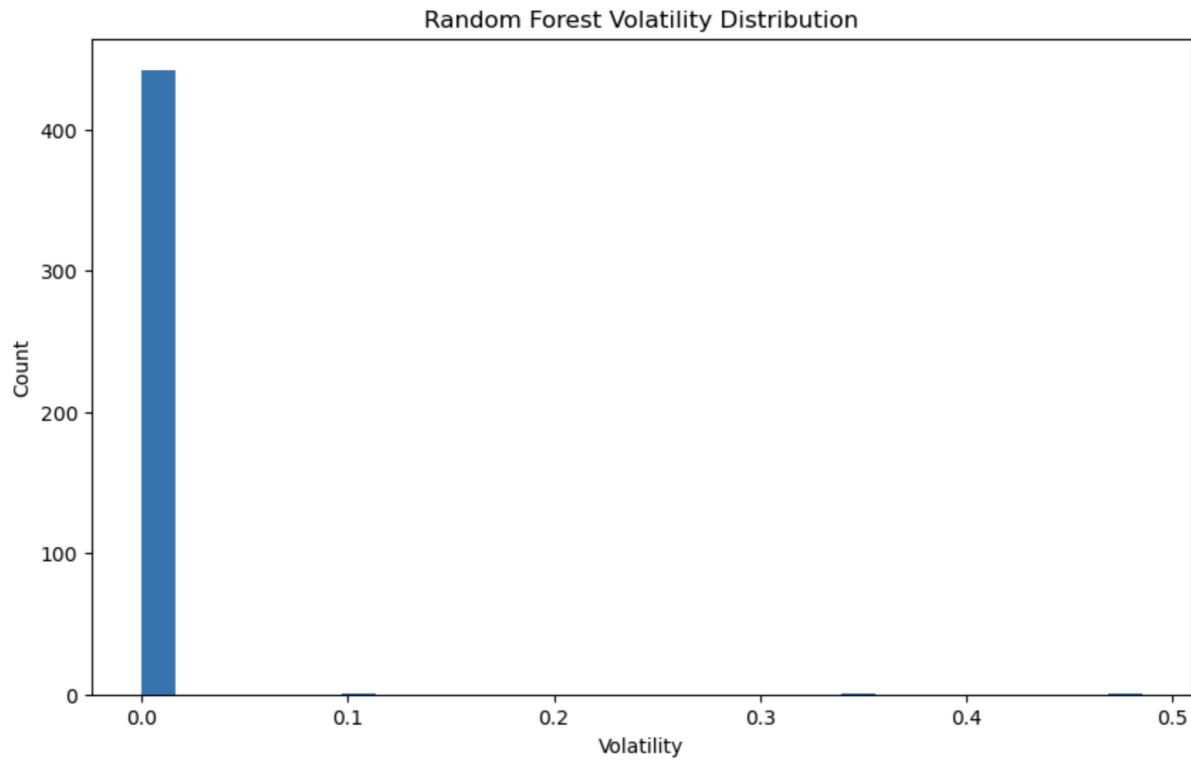


Figure 2

Distribution of Volatility Scores for Random Forest



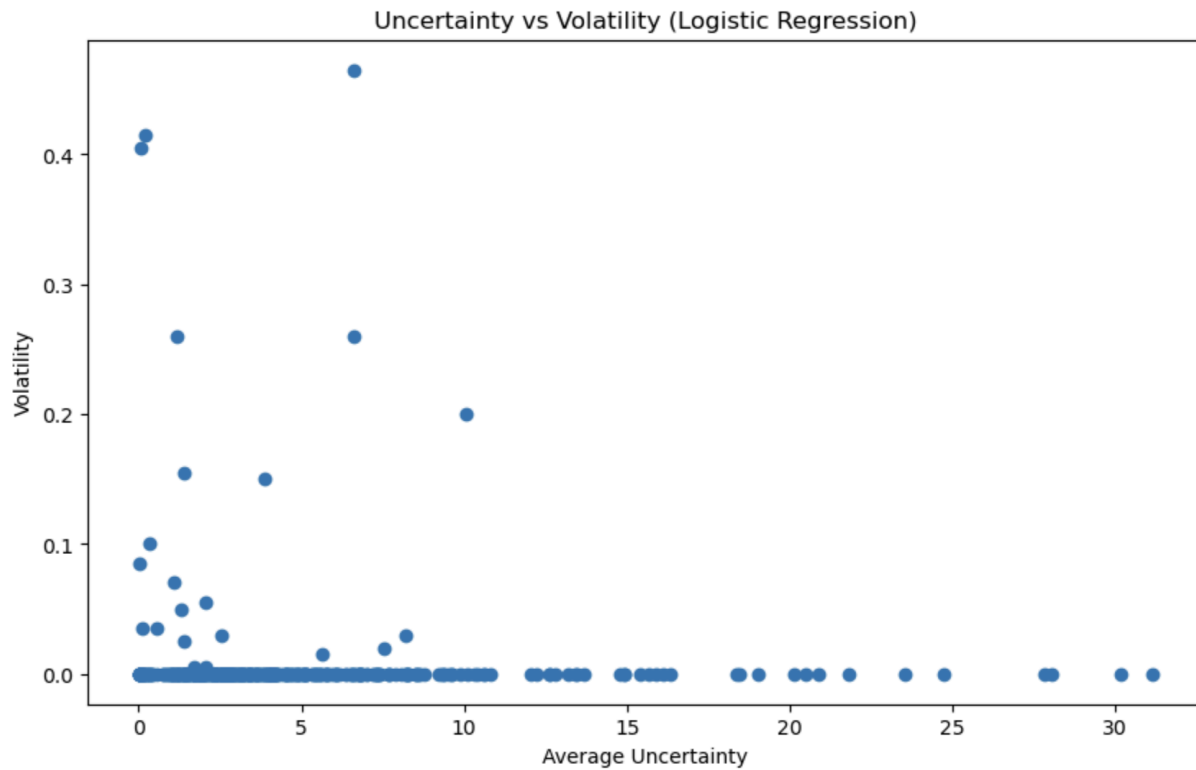
Relationship Between Uncertainty and Volatility

To further examine the role of measurement uncertainty, the relationship between average feature uncertainty and volatility was analyzed. Results indicate that higher levels of uncertainty are generally associated with increased prediction instability across observations, as shown in Figure 3.

This finding supports the central premise of the study: predictions derived from uncertain measurements may appear stable in deterministic settings, but can become highly sensitive when realistic variability is introduced.

Figure 3

Relationship Between Average Feature Uncertainty and Volatility



Note. Higher uncertainty levels are associated with increased prediction instability.

Discussion

The results demonstrate that classification accuracy alone does not fully capture the reliability of habitability predictions. While the random forest model achieved high overall accuracy and strong precision, it exhibited lower recall than logistic regression, indicating that some proxy-habitable planets were not identified. Logistic regression, in contrast, achieved perfect recall but produced a higher number of false positives. This contrast highlights a tradeoff between sensitivity and precision, where logistic regression favors detection of rare cases at the cost of false positives, while random forest provides more conservative but precise predictions.

These differences underscore the importance of evaluating multiple performance metrics when working with highly imbalanced datasets.

Importantly, volatility analysis complements these findings by revealing that prediction stability varies independently of traditional performance metrics. Even models with strong precision or recall may produce unstable predictions for certain observations, particularly those near classification boundaries.

Additionally, volatility analysis revealed that certain planets exhibited significant sensitivity to input perturbations. These cases are particularly important, as they represent observations near the model decision boundary where small changes in measured features can lead to different classification outcomes.

This finding has broader implications for exoplanet research. Planets identified as potentially habitable may warrant further scrutiny if their classifications are highly unstable under realistic measurement uncertainty. Conversely, stable classifications may provide greater confidence in habitability assessments.

Limitations

This study has several limitations. First, the dataset exhibits substantial class imbalance, with very few planets meeting the proxy habitability criteria. As a result, model performance metrics for the positive class are based on a small number of observations and may be sensitive to the specific train-test split.

Second, the proxy habitability definition is based on simplified thresholds for planetary radius and equilibrium temperature and does not account for other factors such as atmospheric composition or stellar radiation variability.

Finally, uncertainty was incorporated using a simplified approximation of measurement variability. While this approach captures key aspects of observational uncertainty, it does not fully represent the complex statistical properties of astrophysical measurements.

Conclusion

This study introduced a volatility-based framework for evaluating the stability of exoplanet habitability classifications under measurement uncertainty. By combining standard classification models with Monte Carlo perturbations, this approach provides a simple and interpretable method for assessing prediction reliability.

The results show that even when models produce confident predictions, some classifications are highly sensitive to small variations in input data. This highlights the importance of incorporating uncertainty into data-driven decision-making processes, particularly in domains where measurements are inherently noisy.

Future work may extend this framework by incorporating more sophisticated uncertainty models or applying the approach to additional classification tasks in astrophysics and beyond.

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